

# **Data Understanding and Measurement Basics**

*A Technical Report on Observations, Variables, and Measurement Fundamentals for Biochemistry, Electronics, and Mechanical Domain Data*

Name: Siddeshwar

Domain: Software / Machine Learning

Task Number: Task 3 — Day 1, Week 2

Repository Link: [https://github.com/blackfang007/Synergy\\_TP](https://github.com/blackfang007/Synergy_TP)

Submission Date: 1 July 2026

# 1. Objective

This report builds the theoretical foundation needed before any dataset from Biochemistry, Electronics, or Mechanical domains can be analyzed using tools such as pandas or visualization libraries. Before writing code, an analyst must be able to describe what a dataset actually represents: what an observation is, how variables are classified, how raw data differs from processed data, why measurement units are attached to every number, and what basic summary statistics communicate. Skipping this stage is what leads to comparisons between quantities that are not actually comparable, or conclusions drawn from a single misleading average.

# 2. Observations and Variables

An observation is a single recorded instance — one row of a dataset — corresponding to one entity measured at one point in time or under one condition. For example, one trial of an enzyme assay, one reading from a circuit at a given load, or one tensile test on a metal sample is a single observation.

A variable is a property that is being measured or recorded across observations — it corresponds to one column of a dataset. Temperature, voltage, and material type are all variables; the specific value recorded for one experimental run is the observation's entry for that variable. The distinction matters because statistical operations such as mean or standard deviation are computed down a variable (column), never across an observation (row), and confusing the two leads to nonsensical summaries.

# 3. Categorical and Numerical Data

Categorical data records a label, group, or class rather than a measurable quantity. It answers "what type" or "which group," and arithmetic operations such as averaging do not apply to it — for example, a material labelled "Type A" or "Type B" cannot be averaged. Categorical data may still be ordered (such as low/medium/high load) or unordered (such as control/heated), and this distinction affects which statistical tests are valid later.

Numerical data records a measured quantity that carries a magnitude and, almost always, a unit. It supports arithmetic operations such as averaging, differencing, and scaling. Numerical data is further split into discrete values (countable, such as number of trials) and continuous values (measured on a scale, such as voltage or pH), which again affects which analysis methods are appropriate.

# 4. Domain Examples

The table below gives one categorical and one numerical example from each of the three domains relevant to this task, along with a short note on why the associated unit or context cannot be ignored.

| Domain       | Categorical Example                 | Numerical Example                                  | Why Unit / Context Matters                                                                                                 |
|--------------|-------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Biochemistry | Sample condition: control or heated | pH, concentration (mM), absorbance (AU)            | A pH of 7 and a voltage of 7 V are numerically identical but physically unrelated; the unit defines what the number means. |
| Electronics  | Load type: low load or high load    | Voltage (V), current (mA), resistance ( $\Omega$ ) | 5 V and 5 mA are both "5" but represent different physical quantities                                                      |

| Domain     | Categorical Example   | Numerical Example                                    | Why Unit / Context Matters                                                                                            |
|------------|-----------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|            |                       |                                                      | and cannot be added or compared directly.                                                                             |
| Mechanical | Material type: A or B | Force (N), strain (dimensionless), displacement (mm) | A value of 120 N describes force, not a score or percentage; dropping the unit strips the number of physical meaning. |

Table 1: Categorical and numerical examples across the three domains.

## 5. Raw Data and Processed Data

Raw data is the data exactly as it was collected from an instrument, sensor, survey, or experiment, before any cleaning, transformation, or calculation has been applied. It may contain noise, missing entries, duplicate rows, or values recorded in inconsistent units. Processed data is raw data that has gone through cleaning and transformation steps — unit conversion, outlier handling, aggregation, normalization — so that it is ready for analysis or modelling. Knowing which stage a dataset is in is essential: applying analysis techniques meant for processed data directly to raw data can produce results that look valid but are built on unresolved errors.

## 6. Measurement Units and Why They Matter

A measurement unit is the defined standard quantity against which a measured value is expressed — volts for voltage, newtons for force, moles per litre for concentration. A number without its unit is incomplete: "120" means nothing on its own, but "120 N" specifies a force of a particular magnitude. Units also carry the scale of measurement (milli-, kilo-, and so on), and analysts must confirm that every value in a variable is recorded on the same scale before combining or comparing them.

## 7. Why Comparing Values Across Units Is Dangerous

Two numbers with the same magnitude but different units describe different physical quantities, and treating them as comparable produces meaningless or misleading conclusions. Comparing a pH reading of 7 to a voltage reading of 7 V, or a resistance of 5  $\Omega$  to a current of 5 mA, ignores that the underlying quantities are not on the same scale or even the same physical dimension. This risk also appears in more subtle forms — for instance, comparing concentration values recorded in mM against others recorded in  $\mu\text{M}$  without converting first, which silently introduces a thousand-fold error. Every comparison must first confirm that units match or have been explicitly converted.

## 8. Basic Summary Statistics

Mean is the arithmetic average of all values in a variable, calculated by summing the values and dividing by the count. Median is the middle value when all observations are sorted in order, splitting the dataset into two equal halves. Minimum and maximum are the smallest and largest recorded values in a variable. Range is the difference between the maximum and minimum, giving a first sense of how spread out the data is.

## 8.1 When the Mean Can Be Misleading

The mean is sensitive to extreme values (outliers) and to skewed distributions. A single unusually high or low reading — caused, for instance, by a sensor fault or a contaminated sample — can pull the mean away from where most of the data actually sits, while the median remains largely unaffected. For this reason, the mean alone should never be reported without also checking the median, the range, and the shape of the distribution, particularly for small sample sizes where one anomalous observation carries disproportionate weight.

## 9. Checks Required Before Analyzing a Dataset

Before any formal analysis begins, a dataset should be inspected for the following:

- Column names and what variable each one actually represents.
- The unit attached to every numerical variable, and whether units are consistent across all rows.
- The data type of each column — whether it is categorical or numerical, and whether numerical columns are stored as text by mistake.
- Missing values, duplicate rows, and placeholder entries (such as "NA" or "-1" used to mean missing).
- The experimental or measurement condition under which each observation was recorded, since combining data from different conditions without labelling them separately can hide real effects.
- Whether the data is raw or already processed, and what processing, if any, has already been applied.

## 10. Questions to Ask Before Analyzing Domain Data from Another Team

When a dataset is received from another team rather than collected directly, additional context is needed before it can be trusted:

1. What does each column represent, and what unit is each numerical column recorded in?
2. Under what conditions (equipment, environment, protocol, calibration) was the data collected?
3. Has any cleaning, filtering, or transformation already been applied to this data?
4. Are there known issues — missing values, faulty readings, sensor drift — that the collecting team is already aware of?
5. Is this the complete dataset, or a subset, and if a subset, how was it selected?
6. What was the original purpose of collecting this data, since that purpose can introduce selection bias relevant to a different analysis goal?

## 11. Conclusion

A dataset is not simply a table of numbers; each value is only meaningful when tied to its variable name, its measurement unit, the condition under which it was recorded, and the domain it comes from. Distinguishing observations from variables, categorical from numerical data, and raw from processed data, together with a disciplined set of pre-analysis checks, is the foundation that must be in place before any statistical summary, visualization, or model is built.

This report has laid out that foundation as a basis for the pandas- and visualization-focused work planned later in this module.

## References

- [1] Khan Academy, "Statistics and Probability," Khan Academy. [Online]. Available: <https://www.khanacademy.org/math/statistics-probability>
- [2] D. Diez, M. Cetinkaya-Rundel, and C. Barr, OpenIntro Statistics, 4th ed. OpenIntro, Inc. [Online]. Available: <https://www.openintro.org/book/os/>
- [3] National Institute of Standards and Technology, "Exploratory Data Analysis," NIST/SEMATECH e-Handbook of Statistical Methods. [Online]. Available: <https://www.itl.nist.gov/div898/handbook/eda/eda.htm>